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Video games and close relations: Attachment and empathy as predictors of children's and adolescents' video game social play and socio-emotional functioning

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ABSTRACT

This cross-sectional study examined the differential associations between attachment patterns and empathy and children's and adolescents' aggressive, prosocial and interpersonal video game play and peer relations. The sample was composed of 1391 Israeli children and adolescents, aged 9–15 who rated their recent video-game involvement in terms of violent, prosocial and interpersonal interactions. Dispositional attachment styles and empathy were measured, as well as social satisfaction, prosociality, peer support, subjective well-being, and aggressive behavior in daily life. The results confirmed the relationship between type of video game involvement, social functioning, and well-being. Prosocial and interpersonal video game play were related to greater social satisfaction, peer support, and prosocial behavior, which led to increased well-being, whereas violent video game play was related to increased school bullying, and lower social satisfaction and prosociality. Secure attachment was related to increased empathic concerns and higher levels of prosocial and interpersonal interactions in video game use. Children with attachment anxiety or avoidance engaged in significantly less prosocial video game behavior. These findings challenge the assumption that violence in video games is inevitably related to negative social consequences and points to the potential of video games to serve basic attachment and prosocial needs that can promote close relations.

1. Introduction

Video games have rapidly become one of the most widespread forms of media used by children and adolescents (Anderson, Gentile, & Buckley, 2007). The accessibility and popularity of video games, coupled with the fact that many video games today contain violence in some form (Anderson & Dill, 2000), have led many to express concerns about the potential negative impact of violent video games on children.

Previous studies have linked violent video games to various negative cognitions and behaviors such as aggressive thoughts and hostility (e.g., Anderson & Bushman, 2002; Carnagey, Anderson, & Bushman, 2007; Gentile & Stone, 2005). However, more recent studies suggest that the notion that children and adolescents who play violent video games will inevitably become more aggressive and hostile may be unwarranted, since there are studies that have failed to find such relationships (e.g., Ferguson, 2015; Furuya-Kanamori & Doi, 2016; Jerabeck & Ferguson, 2013).

Some of the inconsistencies in the literature may be related to changes in current video games, and to the personality characteristics of the video game players themselves. Video games have evolved into complex worlds of immersive storytelling, and interactive experiences in which players often are free to decide on different courses of action which can result in completely different game experiences. Thus, even within the same game, players can reconnoiter and scout for enemies, kill and eliminate other players, but also save the life of other players, share weapons and tactics, and display cooperation and teamwork, especially when they play together in teams or groups.

It should come as no surprise that recent studies have indicated that certain video games have a positive impact on prosocial behavior and can foster helping acts and empathic thoughts (Gentile et al., 2009; Greitemeyer & Osswald, 2009, 2011; Harrington & O'Connell, 2016). Both the social and antisocial sides of the coin are important, but without a broadly based study that includes them both, it is hard to grasp the whole picture of the social effects of video games on children and adolescents.

Aside from the games themselves, the personality characteristics of

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children and adolescents need to be taken into account when examining whether video games have adverse or positive effects on their social behavior (Funk, Buchman, Jenks, & Bechtoldt, 2003). Of the few personality dispositions explored by research on the effects of video games on children, much emphasis has been given to trait aggressiveness; i.e., children who are highly aggressive by nature (Anderson & Huesmann, 2007), whereas the possible moderating role of basic positive personality qualities such as secure attachment and empathy have received relatively little attention. This dearth of research is surprising, not only because of the general interest in individual differences in video game players, but also because these two personality traits are basic developmental propensities that predispose children to aggressive or prosocial behavior.

To address these gaps in the literature, the current study asked 9–15 year olds to report their frequency of gaming and to estimate the amount of violent but also prosocial content and interactions in their current favorite video games. The goal was to investigate the social and emotional consequences of prosocial, interpersonal, and aggressive video game interactions in child and adolescent game players. Correlatively it explored whether children's and adolescents' attachment patterns and trait empathy would predict video game aggressive and/or prosocial play, and whether these dispositions could serve as moderators in the associations between video game play and a range of positive and negative social outcomes.

1.1. Violent and prosocial effects of video games

Video games have become more realistic, vivid, engaging, and violent over the course of time and the public and researchers have become increasingly concerned with their negative influences on children's and adolescents' emotional state and behavior (Anderson & Bushman, 2001; Gentile & Stone, 2005). However, there is also growing interest in the positive effects of video games on children which has led to many new studies investigating whether prosocial video games promote prosocial tendencies (Gentile et al., 2009; Harrington & O'Connell, 2016; Vieira, 2014). Nevertheless, the issue of whether violent or prosocial video games affect players' social and non-social behavior is still hotly debated (for a review, see Greitemeyer & Mügge, 2014).

The most influential theory in the field is the General Learning Model, GLM (Buckley & Anderson, 2006) which posits that the effects of video games on social behavior depend to a considerable extent on the situational aspects of the game (e.g., content, challenges, rewards, enjoyment), and the players' personal characteristics (e.g., personality traits, beliefs, values, and attitudes). The personal variables interact with the game situational variables and impact the individual's affect, arousal, cognitions, and actions related to the video game that has been played (Saleem, Anderson, & Gentile, 2012; Swing, Gentile, & Anderson, 2008).

The GLM also claims that repeated and accumulated exposure to video game content, stories, and dynamics may have prolonged effects on cognitive and emotional constructs such as levels of hostility or empathy, desensitization to others' pain, and (Gentile, Groves, & Gentile, 2014; Greitemeyer, Agthe, Turner, & Gschwendtner, 2012; New York University Press; Netten et al., 2015) normative beliefs about whether aggression is legitimate (Gentile, Groves, & Gentile, 2014). Many studies have indeed shown that children and adolescents who engage in more violent video games display more aggression over time, such as delinquency, bullying behavior, and getting into fights (Anderson et al., 2007; Ferguson & Olson, 2014; Hopf, Huber, & Weiβ, 2008; Möller & Krahé, 2009; Wallenius & Punamäki, 2008). Nevertheless, more recent correlational studies are also beginning to explore the relationships between prosocial video game play and prosocial behavior in real life (e.g., Greitemeyer, Agthe, Turner, & Gschwendtner, 2012; Greitemeyer, Osswald, & Brauer, 2010; Sestir & Bartholow, 2010).

Although this literature lends weight to the General Learning Model's underlying assumptions, the results remain contradictory. Several studies have failed to demonstrate that playing violent video games increases aggressive behavior (e.g., Ferguson, 2015; Furuya-Kanamori & Doi, 2016; Jerabeck & Ferguson, 2013), or that playing prosocial video games increases helping behavior (e.g., Rosenberg, Baughman, & Bailenson, 2013). Critics of the literature claim that video game studies often suffer from weak construct validity of the study measures, poorly matched experimental versus control video games, and failure to control for third variables (i.e., confounding variables such as trait aggression, or delinquency history) (Ferguson, 2010; Ferguson & Kilburn, 2010).

However, not all studies have these limitations, and it is still not fully understood why in some studies the association between video game and violence or prosocial outcomes is either nonexistent or weak. A recent study by Przybylski and Weinstein (2019) noted that some of these studies have not adequately differentiated between social and aggressive content and interactions in video games, which can be violent and prosocial at the same time, and therefore do not properly reflect the multifaceted nature of video games today.

1.2. The mixed world of video games

Current day video games are complex experiences composed of multiple content, activities, and processes that may affect the video game players' internal states and guide their learning experience and behavior (Squire, 2011). As a result of the constant innovation and development in graphic quality, avatar representations, and story lines, the classification of video games as violent or prosocial can be difficult. Many involve both aggressive and prosocial content and interactions with game characters and virtual versions of other game players (Gentile et al., 2009).

It is especially common in the popular massively multiplayer online videogames (MMOGs) that allow players to interact not only with the computer-controlled characters but also with other players' characters via their respective digital avatars (Steinkuehler, 2004, pp. 521–528). Virtual social worlds are loosely structured by open-ended narratives, where game players are mostly free to do what they please as long as they stay within the bounds of the game rules: they can slay ogres, save friends' lives, barter goods, besiege castles, or build relationships of solidarity (Jenkins, 2006).

Traditionally, video games with violent content were not considered prosocial, even if the players' characters helped other players by killing mutual enemies. For example, in multiple studies on the social effects of gaming, violent video game engagement was measured by combining responses to reverse-scored items reflecting non-violent game content (e.g., How often do you help other players in this game?), and items reflecting violent content (e.g., How often do you kill other players in this game) in a single violent gaming construct (e.g., Gentile et al., 2009; Gentile, Li, Khoo, Prot, & Anderson, 2014). However, creating two separate variables, one reflecting prosocial game content and the other reflecting violent content, and assessing their relative contributions to children's behavior, could better reflect aspects of social learning that are not completely captured by a single violence construct.

In the current study, children and adolescents reported the incidence and variability of exposure to both aggressive and prosocial behavior in their favorite video games, to better understand the relationships between this exposure and their real-life social functioning (i.e., prosocial behavior, social satisfaction, peer support, and aggression).

1.3. Psychosocial consequences of the social context of video games

Most research on the social effects of video games has focused on the effects of video game contents, especially violent material (Anderson & Bushman, 2002; Anderson et al., 2007). However, recent work has claimed that the social context of the game can shed light on the influence of a game on players' social learning and behavior (Gentile, 2011). Many contemporary video games are designed for multiplayer and

cooperative modes where players are encouraged to play with others. The social context of the game (i.e., playing alone or with other players either cooperatively or competitively) may alter the effects of its content and could enhance certain social or antisocial effects (Gentile et al., 2009). Specifically, the social effects of video games may depend on both their prosocial and antisocial contents and contexts. This implies that frequent playing of video games in which the game characters and players support and help each other in nonaggressive ways may increase long-term prosocial behaviors, even if the content of the games is violent.

Theoretically, we expected that the prosocial content of the video game or its context (when playing together in teams or groups) would facilitate several types of social learning. In the short-term context, prosocial games and interactions between players may provide positive social models, require social practice, encourage cooperative behavior, and provide immediate feedback or reinforcement for it. As suggested in the GLM (Anderson & Bushman, 2002), these learned aspects may trigger prosocial feelings, thoughts, and behaviors that influence subsequent social behaviors outside the game, such as prosocial behaviors or fewer aggressive behaviors.

If children and adolescents engage in these short-term social behaviors frequently as a result of repeated involvement in MMOGs or prosocial games, several long-term social effects could occur, including changes in (1) prosocial behavior, such as more helping, sharing, cooperation, and provision of social support for friends; (2) aggressive behavior, such as lower levels of hostility or bullying; (3) a sense of peer support (i.e., sense of togetherness and connectedness to the peer group, especially when playing online games with close friends); and (4) social satisfaction (i.e., evaluation of the relationships with peers as fulfilling and gratifying). Some of these effects may simply be the result of repeated exposure to the prosocial content and interactions of video games. More importantly, however, others may arise from the selective reinforcement of social interactions (such as beliefs about the benefits of prosocial and cooperative behaviors that derive from playing video games where teamwork is rewarded), or from playing multiple MMOGs, and video games with similar prosocial content or with a core group of friends.

There is scant empirical evidence on the effects of the prosocial context of video games on children's and adolescents' social functioning and satisfaction. However, several studies on adult players have shown that players of violent video games that involve cooperative play were more likely to exhibit cooperative, prosocial behaviors in and outside the game than those who played competitive or nonviolent video games (Ewoldsen et al., 2012; Ferguson & Garza, 2011; Velez, Mahood, Ewoldsen, & Moyer-Gusé, 2014). Another study indicated that social playing of violent video games (in groups or teams) reduced feelings of hostility and curbed aggressive behaviors in comparison to playing alone (Anderson & Morrow, 1995; Eastin, 2007; Jerabeck & Ferguson, 2013). Likewise, playing violent video games cooperatively decreased players' access to hostile and aggressive thoughts (Schmierbach, 2010; Velez et al., 2014).

Yee (2006) found that three of the strongest motivations for video game play were significantly related to social satisfaction: the desire to establish meaningful relationships with other people, deriving satisfaction from teamwork, and socializing (chatting with others and helping). In a study on high-school students, Durkin and Barber (2002) reported that a considerable proportion of the time spent in video game play was with family members and friends and that video game play was significantly associated with social involvement and family closeness. Similarly, Jansz and Tanis (2007) found that playing a first-person shooter videogame rarely took place in isolation and that the main predictor of playing time in this type of game was a desire for social interactions.

The potential improvement of children's and adolescents' social relationships as a result of video game social involvement may also contribute to their subjective well-being, a term that encompasses the ways people evaluate their lives. Subjective well-being is comprised of a cognitive component related to life satisfaction that involves viewing one's life in a positive manner, and an affective component related to emotional well-being that assesses and adjusts the balance between positive emotions and negative emotions in the everyday life (Diener, Suh, Lucas, & Smith, 1999).

Most works that have examined the potential influence of video game involvement on well-being have centered on problematic engagement. For example, in a longitudinal study, Lemmens et al. found that adolescents' excessive on-line video game play led to feeling of loneliness (Lemmens, Valkenburg, & Peter, 2011). However, little is known about the consequences for wellbeing of gaming among the non-problematic broader playing community of children and adolescents who engage on almost a daily basis in social video game play.

Specifically, this article examines both the direct and indirect relationships between social and prosocial video game involvement and psychological well-being. In psychological studies, social support and connectedness have been found to be correlated with improved subjective well-being (Goswami, 2012; Malecki & Demaray, 2006). It is often suggested that social satisfaction and support are related to well-being because they elicit positive emotions, predictability in life, and a sense of self-worth and belonging. They also function as stress buffers by reinforcing self-efficacy, self-esteem, helping from support providers, and problem solving behaviors (for a review, Chu, Saucier, & Hafner, 2010). Similarly, we posit that when children and adolescents satisfy their needs for connection and belonging in their daily video game use, they feel a greater sense of subjective well-being.

In addition, video game play may have a direct association with children's and adolescents' well-being. Clearly, video games are fun, engaging, and rewarding for children and teens, and as such they elicit positive emotions. Several studies have established a causal association between playing preferred video games and increases in positive emotion and improved mood (e.g., McGonigal, 2011; Russoniello, O'Brien, & Parks, 2009; Ryan, Rigby, & Przybylski, 2006).

Although not restricted to video game research, the importance of the experience of positive affect on a daily basis has been explored in the Broaden-and-Build theory of positive emotions (Fredrickson, 2001), which speculates that the accumulated experiences of positive emotions can stockpile people's psychological resources and trigger upward spirals towards enhanced subjective well-being in the long term. Thus, we suggest that some of the positive emotional experiences elicited during video game play may contribute directly to children's and adolescents' well-being.

1.4. Attachment and empathy as possible indicators of prosocial video game use

Games that are primarily social and rely on diverse social partners provide a significant amount of variability in the video game experiences, depending on the personal attributes of the game players and the type of engagement during game play (Nagle, Wolf, & Riener, 2016). For example, several studies have shown that children who were high on trait aggressiveness and hostility often played video games in a more hostile and aggressive manner (Anderson & Huesmann, 2007; Funk, Buchman, & Germann, 2000; Funk et al., 2003; Lynch, 1999).

However, too much emphasis has been given to aggressiveness and too little to basic positive personality characteristics that are explicitly known to correlate with social functioning during childhood and adolescence. Especially noticeable by their absence are attachment (Bowlby, 1973) and empathy (Hoffman, 2000) which correspond to personality characteristics that predispose children to prosocial or antisocial behavior across situations.

Bowlby's Attachment theory emphasizes the role of early interactions with primary caregivers in influencing the child's socioemotional development, and trustworthiness and responsiveness to other people in the social world later in life (Bowlby, 1973, 1988). Those who experience reliable and responsive care have confidence in the

availability and supportiveness of meaningful others (i.e., friends/peers, parents, family members), and develop secure attachment (Bowlby, 1973; Waters & Cummings, 2000). Conversely, those who experience neglectful or inconsistent care develop a sense of mistrust and insecurity in the world, have more doubts about their self-worth, and form patterns of insecure attachment, which further parses into avoidant attachment (occurs in neglectful caregiving conditions) and anxiety attachment (occurs in inconsistent caregiving conditions) (Brennan, Clark, & Shaver, 1998). This taxonomy of attachment patterns has been operationalized using self-report measures, interviews and laboratory-based observations that classify participants as corresponding to one of the three attachment styles: secure, insecure-avoidant, or insecure-anxious (Shaver & Mikulincer, 2002).

Securely attached children tend to perceive themselves as being good-natured, likable, and friendly, and others as being generally reliable, well intentioned, and trustworthy (Dykas & Cassidy, 2011). Avoidant children tend to deactivate their attachment needs, distance themselves from significant others out of fear of disappointment, have compulsive self-reliance, and are less likely to ask for help from others in difficult situations (Bowlby, 1982; Main & Solomon, 1986). Anxious children tend to hyperactivate their attachment needs, and develop a clinging, over-dependent, and hypervigilant attitude toward significant others and close relationships (Al-Yagon & Mikulincer, 2004).

Securely attached children show more empathy and cooperation in interpersonal relationships than children with an anxious or avoidant style. Anxious children tend to have anxiety symptoms, are more dependent on others, and are easily irritated related to their deficits in emotional regulation. Finally, avoidant children tend to demonstrate higher levels of aggressive behavior, hostility, lack of empathy, and emotional insulation than secure or anxious children (Ainsworth, 1989; Weinfield, Sroufe, Egeland, & Carlson, 1999). Overall, attachment patterns are relatively stable from early childhood to the school age years and adolescence (Fraley, 2002).

Another personality construct linked to children's social behavior is empathy, which is a fundamental aspect of social cognition. Empathy is defined as an emotional response that stems from a comprehension or apprehension of another person's condition or emotional state (Eisenberg, Spinrad, & Sadovsky, 2006). It involves the ability to accurately perceive and share the feelings of other people ((Eisenberg & Fabes, 1998; Losoya & Eisenberg, 2001). Numerous theorists and studies have shown that empathic children are more likely to act in a prosocial way than less empathic peers (Batson, 1991; Eisenberg et al., 2006; Hoffman, 2000). Empathy can direct children's attention to others' needs, and elicit concern for other people in need, pain or distress (Batson, 1991; Eisenberg, 1986). Associations have also been observed between low levels of empathy and aggression and antisocial behavior (Batanova & Loukas, 2014; Lovett & Sheffield, 2007).

Previous studies examining the relationship between video game play and empathy have mainly focused on the impact of violent video game use, and showed that it is related to lower empathy (e.g. Bartholow, Sestir, & Davis, 2005; Funk et al., 2003; Prot et al., 2014; Uhlmann & Swanson, 2004). However, since modern multiplayer games require group play and cooperation, it is plausible that children and adolescents who are high in empathy may be more prosocial, cooperative, altruistic, and attentive to other players' needs than those who are low in empathy. Similarly, attachment styles may also affect social behavior and interactions in the video game environments, such that children and adolescents who are securely attached may exhibit more prosocial and cooperative video game play. The present study was designed to explore these issues.

1.5. The present study

The studies reviewed above suggest that the content of video games (i.e., violent or prosocial video game plot and goals) is not the most influential factor in determining their social effects on video game players; rather, the social context of the games (e.g., cooperative and prosocial play), and the personal attributes of the game players that can shape the type of social engagement and interactions during game play may be more important (Nagle et al., 2016).

To explore the contribution of contextual and personality factors to players' social behavior both within and outside the game context, the present study examined the relationships between children's and adolescents' video game prosocial and violent play and real-life socio-emotional functioning in terms of aggressive behavior, prosocial behavior, social satisfaction, peer support, and subjective well-being. It also examined the associations between children's and adolescents' psychosocial personality dispositions (i.e., attachment and empathy), prosocial tendencies in video game play, and social functioning. For this purpose, children and adolescents were asked to list their current favorite video games that they played the most often, estimate how many hours they spent weekly playing their favorite games, and rate the levels of prosocial and violent content and interactions in their video game play. They also completed attachment-style, trait empathy, social functioning, and well-being scales.

In light of previous findings on the associations between prosocial and cooperative video game play that were reported to lead to decreased aggressive behavior and enhanced prosocial behavior and social closeness in daily life (e.g., Durkin & Barber, 2002; Ewoldsen et al., 2012; Ferguson & Garza, 2011; Gentile et al., 2009; Jerabeck & Ferguson, 2013; Velez et al., 2014), it was hypothesized that:

H1: Video game prosocial interactions will be associated with children's and adolescents' positive social functioning (i.e., lower aggressiveness, and higher prosociality, social satisfaction, and peer support).

Based on the centrality of social functioning in children's and adolescents' subjective well-being (Chu et al., 2010), and the positive associations between video game social play and positive emotional experiences and feelings (e.g., McGonigal, 2011; Russoniello et al., 2009), we also hypothesized that:

H2: Video game prosocial interactions will be positively associated with subjective well-being (better life satisfaction and positive emotions, and fewer negative emotions).

H3: Social functioning will partly mediate the relationship between video game prosocial interaction and children's and adolescents' subjective well-being.

In addition, based on studies that have demonstrated that securely attached and empathic children are more likely to act in a prosocial way than insecure or less empathic peers (Batson, 1991; Eisenberg et al., 2006; Hoffman, 2000; Weinfield et al., 1999) we expected that:

H4: Secure attachment and empathy will be positively correlated with more prosocial video game play, which will lead to an increase in social functioning (i.e., lower aggressiveness, and higher prosociality, social satisfaction, and peer support). Specifically, children and adolescents with a secure attachment style should be less aggressive and more prosocial and cooperative than those with an insecure attachment styles (i.e., anxious or avoidant styles) both inside and outside the game environment.

Finally, in view of the well-known relationship between secure attachment and the development of empathy in children (Thompson & Gullone, 2008) we also assumed that:

H5: Empathy should mediate the association between secure attachment and children's and adolescents' prosocial behaviors in video games.

2. Method

2.1. Participants

A random sample of 1426 fourth to tenth graders representative with respect to age, gender and socioeconomic status (SES), was recruited from three elementary and four middle schools in two small cities in the center of Israel, with the assistance of the municipal Education

Departments. Inclusion criteria included playing video game at least 7 h a week. Thirty-five participants that did not fill out all the scales were excluded from the analyses. Therefore, the final sample consisted of 1391 9–15 year olds (710 boys and 681 girls; mean age = 11.97, *SD* = 2.01). The majority came from middle SES homes (63%) and the rest from low SES (18.9%) and high SES (18.1%) homes. Students were mostly Jewish (97%), with 8% reporting orthodox observance, 21% traditional and 71% secular. According to the participants' frequency reports 39.6% played video games almost every day, 30.6% played four or five times a week, 20.5% played two or three times a week, 7.9% played once a week, and 1.4% played a several times a month. The average daily playtime was 2.23 h (*SD* = 1.11).

2.2. Measures

Violent and pro-social behavior in video games. Participants listed their three most recent favorite video games (i.e., the games played most often in the previous month), estimated the number of hours spent per week playing each video game, and rated the degree of violent and prosocial content and interactions in each game. They then completed six items related to prosocial content and interactions with other players or game characters in the three games (i.e., how often the respondent's avatar does nice things for or to other players, shares resources with other players, helps others, cooperates with others, saves the life of others, takes risks to help others), and six items related to violent content and interactions (e.g., how often the respondent's avatar kills, hurts others, steals, causes damage). These items were developed for the present study. The items were scored on a 5-point scale ranging from "never" to "always". Prosocial and violent interactions in each game were multiplied by the amount of time (in hours) playing that video game and averaged across the three games (Anderson et al., 2007). To help the respondents choose their favorite games, participants were provided with a list of video games. This game list, which we compiled, was composed of the names of the most popular video games in 2019 for children and adolescents.

Social interactions and communication with other players. Participants reported the frequency of playing and communicating with different types of co-players in their three favorite games using six items developed by Meng, Williams, and Shen (2015). Respondents were asked to indicate how often they played with each of the following: a family member, a friend whom they were familiar with offline, and a friend they met online. In addition, they reported how often they talked or communicated with other players through each of the following platforms: voice chat, in-game text chat, and face-to-face (e.g., playing in the same room). Items were rated on a 5- point Likert scale ranging from 1 = rarely to 5 = always. Social interactions were multiplied by the amount of time playing each video game and averaged across the three games.

Social Satisfaction Questionnaire. An abridged form of the Loneliness and Social Dissatisfaction Questionnaire (Asher, Hymel, and Renshaw's, 1984; Jarvinen & Nicholls, 1996) was used as a measure of satisfaction in peer relationships. This questionnaire is composed of six items that focus on feelings of loneliness and children's/teens' appraisal of their current peer relationships (e.g., "I am lonely at school" "I have some really good friends", "It's hard to me to make friends at school"). Participants rated each item on a 5-point Likert scale. Total scores could range from 6 to 30, with greater scores indicating greater social satisfaction. This questionnaire is reported to have good internal consistency ($\alpha > 0.90$) in studies on elementary and high school students (Asher, Parkhurst, Hymel, & Williams, 1990). The internal reliability in this study was comparable ($\alpha = 0.91$).

Perceived Peer Support. Participants completed a 4-item subscale from the Multidimensional Scale of Perceived Social Support (Zimet, Dahlem, Zimet, & Farley, 1988) that measures children's/teens' perceptions of their peers as providers of support. The items were "I can talk about my problems with my friends", "My friends really try to help me," "I have friends with whom I can share my joys and sorrows", and "I can count on my friends when things go wrong." Items were rated on a scale from 1 = strongly disagree to 5 = strongly agree and were averaged to form a peer support score. The scale had a good internal consistency in this study ($\alpha = 0.84$).

Prosocial behavior. The prosocial subscale of the Strengths and Difficulties Questionnaire (Goodman, 1997) was used as a measure of children's and adolescents' prosocial behavior. This 5-item self-report scale assesses helpfulness and consideration towards peers (sample items: "I usually share with others, for example CD's, games, food"; " I try to be nice to other people. I care about their feelings"). Degree of prosocial behavior was assessed on a 3-point Likert scale from "not true" to "certainly true." The individual scores for all five items were summed to generate a prosocial score. The internal consistency (Cronbach's alpha) of the scale in the present study was 0.80.

Aggressive behavior. Participants completed an 11-item bullying questionnaire (Sourander et al., 2010) that measures involvement in peer bullying over the previous six months. Participants were asked to indicate how often in the last six months they were involved in aggressive behavior towards others in school, such as slapping or pushing someone, spreading rumors, gossiping, mocking or ignoring someone on a 4 -point scale ranging from Never; Less than once a week; More than once a week, and Most days. Previous research reported good reliability for this questionnaire (Klomek et al., 2011). The Cronbach alpha for the scale in this study was $\alpha = 0.87$.

Life Satisfaction. The Brief Multidimensional Students' Life Satisfaction Scale (BMSLSS) (Seligson, Huebner, & Valois, 2005; Sourander et al., 2010) was used to examine the cognitive component of subjective well-being (Diener et al., 1999). The BMSLSS is a five-item self-report measure for children and adolescents that evaluates life satisfaction in different life domains. Items are rated on a 7-point scale ranging from 1 (terrible) to 7 (delighted). Each item measures life satisfaction in one of five different areas (e.g., "I would describe my satisfaction with my friendships/family life/school experience/where I live/myself as …"). The items are summed to create a general life satisfaction score. Previous studies support the high reliability of the scale (e.g., Shoshani & Slone, 2017). The scale had good internal consistency in this study ($\alpha = 0.82$).

Emotional well-being. The Positive and Negative Affect Schedule for Children (PANAS-C; Ebesutani et al., 2012) was used to assess the emotional component of subjective well-being. This scale is designed for children and adolescents and measures levels of negative and positive affect over the previous few weeks. The PANAS-C consists of 10 items that present five positive emotions (e.g., enthusiastic, proud, excited) and five negative emotions (afraid, upset, nervous). Responses are rated on a 5-point scale ranging from "very slightly or not at all" to "extremely". The alpha coefficients in this study were 0.83 and 0.78 for the positive and negative emotions subscales respectively.

Attachment style. Participants' attachment style was measured using the Attachment Style Classification Questionnaire (Finzi, Cohen, Sapir, & Weizman, 2000). This questionnaire is made up of 15 items measuring three factors, which tap Ainsworth's three classifications of attachment styles: secure (e.g. "I make friends with other children easily"), insecure-anxious (e.g. "Sometimes I'm afraid that other kids won't want to be with me"), and insecure-avoidant (e.g. "Sometimes others get too friendly and too close to me"). Participants rated the extent to which the items described themselves on a 5-point Likert scale ranging from "not at all" to "very much". The ASCQ has demonstrated high reliability and concurrent validity in previous studies (Finzi et al., 2000; Finzi, Har-Even, Weizman, Tyano, & Shnit, 1996), and good reliability for the subscales in this study (Cronbach's $\alpha = 0.81$- 0.86).

Empathy. The Empathy Questionnaire for Children and Adolescents (Netten et al., 2015) was used to assess participants' empathy skills. This questionnaire is composed of 18 items, divided into three subscales (1) affective empathy (e.g., 'If someone in my family is sad, I feel really bad'): measuring the extent to which the respondent feels for the emotions of others (2) cognitive empathy (e.g., 'If a friend is sad, I understand mostly why'), measuring the extent to which the respondent

understands why other people are in distress, and (3) intention to comfort (e.g., 'If a friend has an argument, I try to help') measuring the extent to which the respondent is inclined to help or support other people in need. These items were rated on a 3-point Likert scale ranging from "not true" to "true" and were summed to a total empathy score. The questionnaire had a good internal consistency in this study (α = 0.84).

2.3. Procedure

After receiving academic and municipal ethics committee authorizations and consent from the school principals, the schools obtained written informed consent from the students and their parents. Sixty-four students (or their parents) refused to participate. Data were collected in May 2019. The students completed the questionnaires in school using the Qualtrics Online Survey System in the presence of two research assistants, who were undergraduate students in psychology. The assessments were administered in school classrooms during the school day. Students were informed that participation was voluntary and that their responses would be anonymous and confidential.

2.4. Data analysis

Descriptive statistics and Pearson correlations were used to characterize the sample, and assess the associations between variables. Independent t-tests were conducted to explore gender and age differences across the variables. MANOVA analyses were conducted to examine the effects of the three attachment styles on the outcome scores. A Bonferroni correction was applied for multiple tests.

Structural equation modeling (SEM) was employed to determine the relationships between violent, interpersonal and pro-social interactions in video games, and participants' social functioning and subjective well-being (see Fig. 1). A second model examined the relationship between secure attachment and empathy and violent and prosocial video game play and social functioning in daily life (see Fig. 2). The parameters of these models were examined with maximum likelihood (ML) estimation and bootstrapping using AMOS 25.0. We generated 5000 bootstrap samples to derive less biased standard errors and 95% confidence interval (CI) bootstrap percentiles. We report two types of fit indices: Non-Normed Fit Index (NNFI) and Comparative Fit Index (CFI), and two indices of misfit: Root Mean-Square Error of Approximation (RMSEA) and Standardized Root Mean-Square Residual (SRMR). A NNFI and CFI close to or above 0.90, combined with RMSEA below 0.08 and SRMR below 0.10, are indicative of acceptable fit.

3. Results

3.1. Descriptive statistics

The means, standard deviations, and bivariate correlations for the variables are reported in Table 1. Skewness and kurtosis were consistent with normal distributions. Video game violent interactions were positively correlated with aggressive behavior and negatively correlated with prosocial behavior, social satisfaction, and well-being. Both video game interpersonal interactions and prosocial behaviors were positively correlated with prosocial behavior, peer support, social satisfaction, and subjective well-being. Secure attachment and empathy were positively associated with video game interpersonal interactions and prosocial behaviors. In addition, empathy was negatively associated with video game violent behaviors.

Independent *t*-test analyses were conducted to explore gender and age differences across the variables. The Bonferroni correction was applied, taking into account the number of comparisons. The level of significance considered was $p = .005$. The two developmental age groups tested were pre-adolescence (9–11.9) and adolescence (12–15). Boys reported significantly higher levels of violent video game playing, aggressive behavior, and video game interpersonal interactions than girls, whereas girls reported significantly higher levels of prosocial behavior, peer support, and empathy than boys. Adolescents had significantly higher levels of prosocial video game playing, and prosocial behavior in daily life, but lower subjective well-being than pre-adolescents. The means, standard, deviations, and univariate comparisons are shown in Table 2.

3.2. Video game interactions, social functioning and subjective well-being

A structural equation model using Amos 25 software was conducted with video game interpersonal, prosocial, and violent interactions predicting participants' social functioning and subjective well-being. An interaction between video game prosocial and violent behavior was also investigated but failed to reach statistical significance and did not yield the best goodness-of-fit. The final model consisted of one measured variable of video game violent behavior, and three latent variables:

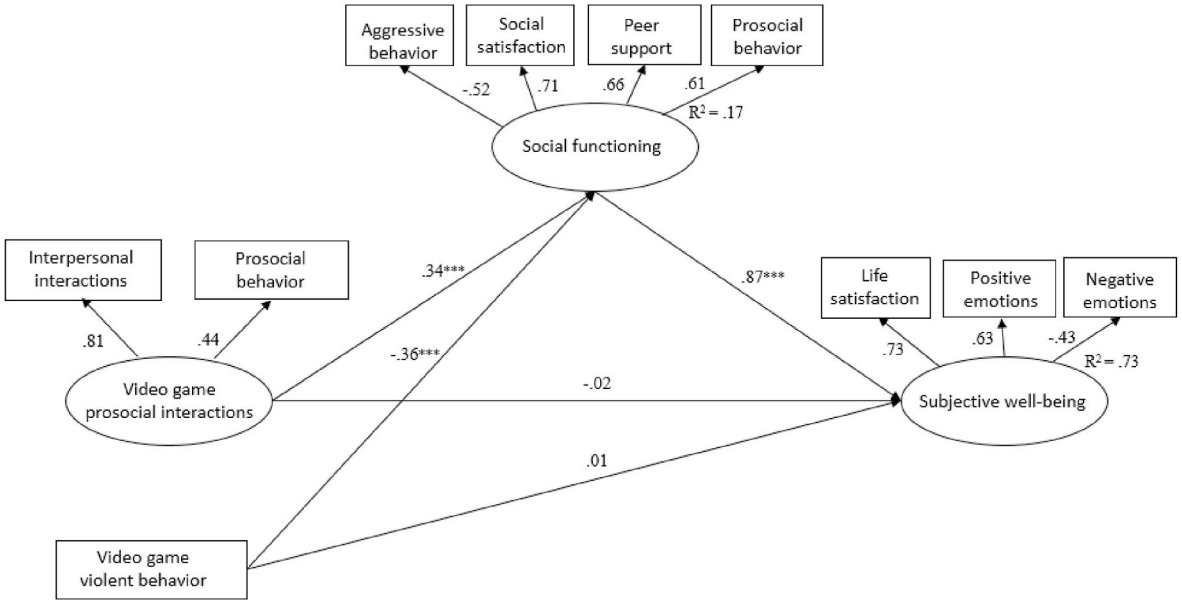


Fig. 1. Video game prosocial and violent interactions, social functioning and subjective well-being. All estimates are fully standardized. ***$p < .001$.

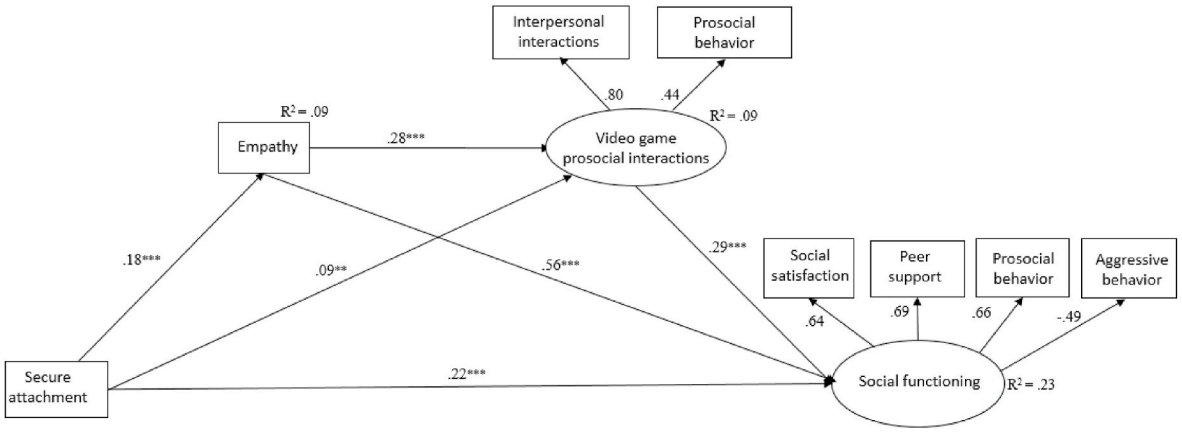


Fig. 2. Secure attachment, empathy, video game social interactions, and social functioning. All estimates are fully standardized. ***$p < .001$, **$p < .01$.

Table 1
Means, standard deviations, and correlations.

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Age	11.97	2.00													
2. Gender (boys)			.03												
3. Video game violent behavior	5.14	6.56	.06*	.26***											
4. Video game interpersonal interactions	3.53	1.30	.01	.14***	.31***										
5. Video game prosocial behavior	4.04	2.48	.14***	-.01	.04	.35***									
6. Prosocial behavior	8.35	1.84	-.11***	-.18***	-.27***	.08**	.29***								
7. Social satisfaction	24.67	4.50	.03	-.02	-.19***	.08**	.17***	.36***							
8. Peer support	3.81	1.01	-.04	-.14***	-.09***	.15***	.19***	.38***	.54***						
9. Aggressive behavior	1.29	0.55	.04	.13***	.40***	.10***	-.09**	-.44***	-.37***	-.24***					
10. Life satisfaction	32.33	6.04	-.11***	-.03	-.17***	.06*	.13***	.36***	.49***	.43***	-.28***				
11. Positive emotions	19.20	3.79	-.13***	-.11***	-.12***	.11***	.17***	.40***	.38***	.42***	-.22***	.45***			
12. Negative emotions	9.26	3.62	.03	-.14***	.08***	-.02	-.08***	-.10***	-.34***	-.20***	20***	-.35***	-.25***		
13. Attachment (secure)			-.04	.03	-.01	.08***	.11***	.17***	.27***	.26***	-.14***	.35***	.24***	-.19***	
14. Empathy	32.80	5.01	-.01	-.24***	-.17***	.10***	.22***	.48***	.28***	.44***	-.23***	.27***	.31***	-.01	.17***

Note. * indicates $p < .05$. ** indicates $p < .01$. *** indicates $p < .001$. $N = 1391$.

Table 2
Gender and age differences for the main variables.

	Gender						Age group					
	Boys (n = 681)		Girls (n = 710)				Preadolescents (n = 612)		Adolescents (n = 779)			
Dependent Variable	M	SD	M	SD	*t*	*P*	M	SD	M	SD	*t*	*p*
Video game violent behavior	6.81	7.45	3.41	4.94	−9.98***	<.001	4.75	6.22	5.47	6.82	−2.05	.04
Video game interpersonal interactions	3.71	1.32	3.37	1.23	−5.14***	<.001	3.53	1.23	3.55	1.34	−0.19	.85
Video game prosocial behavior	4.02	2.29	4.06	2.66	0.21	.83	3.76	1.65	4.25	2.95	−3.66***	<.001
Prosocial behavior	8.01	1.93	8.69	1.68	6.92***	<.001	8.53	1.68	8.20	1.95	3.38**	.001
Social satisfaction	24.59	4.54	24.74	4.46	0.65	.51	24.58	4.62	24.75	4.40	−0.76	.45
Peer support	3.67	1.03	3.96	.970	5.36***	<.001	3.85	1.00	3.79	1.02	1.19	.23
Aggressive behavior	1.36	0.61	1.22	0.45	−5.03***	<.001	1.28	0.54	1.30	.540	−0.88	.38
Life satisfaction	32.14	6.22	32.53	5.85	1.22	.22	32.98	5.89	31.83	6.12	3.52***	<.001
Positive emotions	18.80	3.95	19.63	3.56	4.12***	<.001	19.72	3.53	18.80	3.94	4.55***	<.001
Negative emotions	8.76	3.53	9.77	3.65	5.26***	<.001	9.09	3.42	9.39	3.77	−1.51	.13
Empathy	31.60	5.03	34.04	4.69	9.35***	<.001	32.82	4.91	32.78	5.10	0.12	.90

Note. **$p < .01$. *** indicates $p < .001$. $N = 1391$.

"video game social interactions"; i.e., the interpersonal interactions, and prosocial behaviors in the most recently played three video games, "social functioning"; i.e., participants' social satisfaction, perceived peer support, and prosocial behavior, and "subjective well-being"; i.e., participants' life satisfaction, positive emotions, and negative emotions. The model estimated the direct and indirect effects of the different types of video game content and interactions on participants' social functioning in daily life that served as the mediator, and their subjective well-being. The standardized estimates and multiple squared correlations are presented in Fig. 1.

A multi-group model was used to test for structural invariance as a function of gender and the age of the participants (pre-adolescence: 9–11.9; adolescence: 12–15). The results indicated minimal differences between genders and age groups, suggesting that the model described in Fig. 1 was configurally invariant across genders and age groups, and was run as a single group, though gender was included in the model as a covariate. However, for clarity, these paths are not shown in Fig. 1.

The model for the entire sample was adequate, χ^2 (27) = 344.81, $p <$ .001, NNFI = 0.90, CFI = 0.91, SRMR = 0.06, RMSEA = 0.06. There was a significant direct effect of video game social interactions on participants' social functioning ($B = 0.34$; $SE = 0.19$, p < .001), confirming H1. Video game violent behavior was related to lower levels of social functioning (i.e., lower social satisfaction, peer support, and prosocial behavior; $B = -0.36$; $SE = 0.02$, $p < .001$). Indirect effects were examined using 5000 bootstrapping samples. As predicted in H3, social functioning significantly mediated the relationship between video game social interactions and subjective well-being ($B = .29$; $SE = 0.06$, 95% CI = 0.18–0.41, $p < .001$). Furthermore, the indirect effect of video game violent behavior on subjective well-being through the path of social functioning ($B = -0.22$; $SE = 0.04$, 95% CI = −0.15 to −0.29, $p < .001$) was also significant. However, contrary to our expectations in H2 and H3, the direct effects linking video game violence ($B = 0.01$; $SE = 0.04$, $p = .83$) or video game social interactions ($B = -0.02$; $SE = 0.04$, $p = .60$) with subjective well-being were not significant after controlling for social functioning, suggesting that social functioning fully mediated the associations between video game exposure and well-being.

3.3. Secure attachment, empathy, and video game social interactions

To explore the effects of attachment styles on the respondents' video game use and social behavior a one-way MANOVA was performed with attachment style (secure/anxious/avoidant) as the independent variable, and the outcome scores (video game violent, prosocial, and interpersonal behavior, aggressive behavior, social functioning, and subjective well-being) as the dependent variables. The MANOVA yielded a significant effect for attachment styles on the outcome scores, Wilks' $\Lambda = 0.83$, F (22, 2758) = 12.44, $p < .001$, partial $\eta 2 = 0.09$. Follow-up ANOVA analyses indicated that secure respondents did not differ from insecurely attached respondents (i.e., anxious or avoidant) in levels of violent video-game behaviors, F (2, 1388) = 1.12, $p = .33$, partial $\eta 2 = 0.01$, but had significantly higher levels of social interactions with other players, F (2, 1388) = 22.43, $p < .001$, partial $\eta 2 = 0.03$, and video game prosocial behaviors, F (2, 1388) = 12.73, $p < .001$, partial $\eta 2 = 0.02$. Tukey's pairwise comparisons (with a Bonferroni correction for multiple comparisons) showed that securely attached and insecure-anxious attached respondents played more with other partners than avoidant respondents (*ps* < .001). In addition, securely attached respondents scored higher on video game prosocial behavior than anxious or avoidant attached respondents (*ps* < .001). The securely attached participants reported the lowest levels of real-life aggressive behavior and the greatest social satisfaction, peer support, prosocial behavior, subjective well-being, and empathy, compared to the insecurely attached participants (*ps* < .001).

A structural equation model was conducted to test the relationship between participants' secure attachment and empathy with their social interactions (interpersonal relationships and prosocial behavior) in video game play and their predictions of adolescents' social functioning in daily life. The relationships between secure attachment and video game violent behavior failed to reach statistical significance and did not yield the best goodness-of-fit and hence was not included in the final model (see Fig. 2). Each path in the final model was tested as a function of gender or age group (pre-adolescence and adolescence), but no gender or age differences were found, and the model was run with both genders and age groups combined, with gender as a covariate. For simplicity, the control paths are not shown in the figure. The model had an acceptable fit, χ^2 (15) = 186.97, $p < .001$, NNFI = 0.92, CFI = 0.93, SRMR = 0.05, RMSEA = 0.08 (Fig. 2). As posited in H4, secure attachment *($B = 0.09$; $SE = 0.04$, $p = .01$)* and empathy ($B = 0.28$; $SE = 0.01$, $p <$ .001) predicted higher levels of video game social interactions.

The indirect effect of secure attachment on video game social interactions through the path of empathy (B = 0.05; SE = 0.01, 95% CI = 0.03–0.07, p < .001) was significant, thus, confirming H5. The analyses indicated partial mediation, where the direct effect of secure attachment on video game social interactions remained significant even after accounting for empathy *($B = 0.09$; $SE = 0.03$, $p = .01$)*. The total indirect effect of secure attachment on social functioning was significant (B = 0.14, SE = 0.02, 95% CI = 0.10–0.18, p < .001), indicating that empathy and video game social interactions significantly mediated the relationship between secure attachment and social functioning.

4. Discussion

The current study explored the associations between video game prosocial and violent play and children's and adolescents' social functioning and subjective well-being through the prism of empathy and attachment theory. Examining the foundational roles of attachment and empathy was designed further unravel the relationships between teen and preteen psychosocial personality dispositions and video game involvement and help determine why children and adolescents are increasingly involved in specific video games, how they utilize this medium, and what they gain or lose socially as a result. However, before the roles of attachment and empathy can be discussed, the connections between video game play and social behavior first needed to be confirmed.

As predicted, prosocial video game behavior and playing with co-players or in teams were significantly related to increases in social satisfaction, perceived peer support and prosocial behavior in real-life settings. In contrast, video game violent behavior was related to higher school bullying and lower levels of social satisfaction and prosocial behavior. These findings are consistent with the predictions based on the General Learning Model, and the extensive video game social research literature (Anderson & Bushman, 2001; Buckley & Anderson, 2006; Gentile et al., 2009). While it is impossible to infer causation from the wide range of video-game environments mentioned in this study, many of the respondents' favorite video games were multiplayer games that encourage players to cooperate with other players and contribute to their team in order to win. In this type of game, the players often learn to listen to the suggestions of others, distribute tasks and responsibilities based on skills, and formulate plans and tactics with other players (Cole & Griffiths, 2007). Thus, the shared experience of the social video games presents the player with opportunities to relate to others as in the real social world, and to practice prosocial skills such as helping, reciprocating, cooperating, and sharing that can foster prosocial behavior in day-to-day life.

Beyond the prosocial effects, the findings also indicated significant associations between interpersonal and prosocial videogame play, and feelings of social support and social satisfaction. Events in which the individual is acknowledged, supported by others, or able to provide help to others are all interpersonal experiences that promote a sense of relatedness (Weinstein & Ryan, 2010). Relatedness needs are fulfilled when players connect with other players in a way that make them feel that they belong and matter to others (Martela & Ryan, 2016; Ryan &

Rigby, 2020). Children and adolescents may derive meaningful relationships from video game worlds, and form supportive social networks and strong friendships, both of which are important aspects of young people's psychological well-being (Goswami, 2012).

These findings extend previous work by exploring the simultaneous impacts of the violent and prosocial content of multiple video games, rather than examining the impact of video games that have been classified as violent or prosocial. It is probably difficult to classify many of the current popular video games as either "good" or "bad." What emerges from this study, as in many others, is that video games are powerful and influential teachers that affect children's and adolescents' social behavior in multiple ways and dimensions. Some of their social effects can be beneficial (e.g., the effects of prosocial content on prosocial behavior and sense of social support). Other effects can be harmful (e.g., the effects of violent video game content on aggressive behavior).

Thus, video game influences are complex and may be better understood as multiple domains rather than a simple "good-bad" dichotomy. At the same time, only main effects were found, not interaction effects for social and violent video-game content on children's social behavior. Therefore, further studies are needed to determine whether prosocial aspects buffer the detrimental effects of the violent aspects of these games, and to better explore the interactive dynamics of different aspects of video game environments that may translate into different outcomes for different individuals.

The findings on the differences in video-game prosocial play between children and adolescents with secure and insecure attachment patterns substantiated the hypothesis that secure attachment predisposes adolescents and preadolescents to prosocial behavior and interactions not just in real life settings but also in the social environment of video games. In this study, children who were high on attachment avoidance reported a significantly lower frequency of playing with other players such as family members, friends or children they met online. In addition, they reported a lower level of prosocial play compared to the securely attached respondents. These results are typical of avoidant attachment patterns, which are characterized by the avoidance of close relationships, high self-sufficiency, and the dismissal of interpersonal relationships driven by the belief that they will be disappointed in the end (Weinfield et al., 1999).

In addition, the findings identified both secure and anxious attachment as motivators for multiplayer online games. However, respondents high on attachment anxiety showed significantly less prosocial video game behavior. Thus, unlike securely attached respondents, those with high anxious attachment seemed to seek out social interactions with other players but were less likely to manifest prosocial behavior towards those game partners. Anxious attachment has been found to be related to heightened personal distress and less empathic concern when witnessing others' difficulties and distress (Shaver, Mikulincer, & Cassidy, 2019). Since a wide variety of video games can potentially produce social satisfaction for players high in attachment anxiety, due to their intense dependency needs and emotion regulation difficulties, the prosocial potential of video games may not be fully realized in anxious children and adolescents, as compared to those who are securely attached, who are more able to empathize with other people and relieve others' distress.

It is worth nothing there were no differences in violent video game involvement between the three attachment patterns. This may be due to the fact that in many current video games, violent acts are not a choice that players make but rather are part of the rules of the game that the players need to follow to play effectively and achieve optimal results. However, the prosocial aspects of these games (e.g., communication with other players, sharing, and helping) seem to be more self-directed and autonomous and thus may allow for more of the players' personalities to be expressed in the game. Future studies should include level of choice in prosocial or violent actions as part of the methodology to explore this issue.

The results also indicated that empathy partially mediated the associations between secure attachment and video game prosocial interactions. In other words, securely attached children and adolescents had increased empathic concerns that were related to higher levels of prosocial and interpersonal interactions in their video game use. The combination of secure attachment with empathy, as predicted, appeared to be associated with higher quality interactions with other video game players, resulting in greater benefits from video-game involvement.

For this reason, the issue of video game social effects should be approached from a risk and protective factors perspective. Clearly, video game violence is not necessarily the cause of aggression. When children and adolescents have multiple risk factors for aggression and violence, as shown in the past, they are more likely to display aggressive behavior when they expose themselves to video game violence (Gentile & Sesma, 2003). However, when children and adolescents have protective factors such as empathy and secure attachment, as found in this study, or positive attitudes, helpful intentions, and many other factors that are worth investigating in the future, these attributes may amplify prosocial behavior and learning from video game play.

Thus overall, the present data lend support to the hypothesis that video game influences operate in both directions: players are affected by aspects of the games they choose to play, but they also influence the game environment through their own traits that lead them to seek specific kinds of interactions with others.

4.1. Limitations

While the present study contributes to a better understanding of the associations between attachment, empathy, video game involvement, and socio-emotional functioning, there are several limitations that should be noted. First, in this study the respondents referred to a wide variety of video games that probably differed not only in their level of prosocial engagement or violence but also in terms of enjoyment, plot structure, degree of difficulty, type of violence, and level of autonomy. As a result, the differences in social functioning or well-being cannot be attributed solely to the effects of prosocial and violent content and interactions. The lack of control of such variables represents a limitation that further studies should take into account. This study was also limited by its correlational design; strong inferences about causal relationships between the variables cannot be made. Longitudinal studies are needed to track social behavior changes over time and to evaluate the impact of changes in video game preferences and involvement on children's and adolescents' social functioning and well-being.

In addition, this study relied on self-report measures of social and emotional functioning. Although the measures employed, such as the attachment or empathy questionnaires, have been shown to have high external validity in previous studies, reports about social functioning and school aggressive behavior are sensitive to socially desirable responding. This study was also susceptible to demand characteristics; e.g., the participants' possible awareness of the study goals could have resulted in a tendency to respond consistently or in a socially desirable manner. Thus, future studies should consider including both objective and subjective measures of attachment and social skills when evaluating their relationships with video game involvement.

5. Conclusion

Despite the limitations of this study, its findings are highly relevant to the longstanding debate on the positive and negative effects of video games on children and adolescents. The positive relationships between attachment, empathy, and prosocial play challenge the prevalent assumption that video game play is inevitably related to negative socio-emotional outcomes by suggesting that video games may provide opportunities to practice empathy, compassion, solidarity, and other forms of care and support for others. The results also highlight the potential for video game spaces to serve basic attachment needs by offering a social platform that can promote a sense of belonging, security, and closeness

for those with secure attachment or a high need for peer attachment. Taken together, these findings question the widespread dichotomous view of video games as all bad or good and show that video games may be socially and emotionally impactful at the personal level in both negative and positive ways. In this way, video game play can be seen as a social phenomenon that is situated in a social environment that enables children and adolescents to express and extend their socio-emotional capabilities like any other social context.

These conclusions have clear implications for the type and level of parental involvement in children's and adolescents' video game exposure. Parents often set limits on the amount of children's video game use, and worry about the excessive amount of time their children spend on video game play (Gentile & Walsh, 2002). However, parents' involvement on the content and the social characteristics of the games can be a powerful protective or enhancing factor for their children's social learning and development. It is important that parents know which games their children are most involved in, understand the reasons why they spend time in specific games, be familiar with their positive or negative learning outcomes, and encourage them to engage in good, prosocial, and responsible video game use that fits their personality and psychological needs. In this way, rather than focusing entirely on the negative potential outcomes, video game engagement can be framed in terms of new learning possibilities and as a mechanism for social growth and positive development in children and adolescents.

CRediT authorship contribution statement

Anat Shoshani: Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Supervision. **Shahar Braverman:** Project administration, Investigation, Writing - review & editing. **Galya Meirow:** Project administration, Investigation, Writing - review & editing.

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